# MH1101 Tutorial 8 (Week 9)

**Reference:** Sections 4.3, 4.4 (Lecture Notes)

**1.** Find the limit of the sequence:

$$\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \ldots$$

**2.** Show that the sequence defined by

$$a_1 = 1, \quad a_{n+1} = 3 - \frac{1}{a_n}$$

is increasing and $0 < a_n < 3$ for all $n$. Deduce that the sequence $\{a_n\}$ is convergent and find its limit.

**3.** Show that the sequence defined by

$$a_1 = 2, \quad a_{n+1} = \frac{1}{3 - a_n}$$

satisfies $0 < a_n \leq 2$, and is decreasing. Deduce that the sequence $\{a_n\}$ is convergent and find its limit.

**4.** Define $e_n = \left(1 + \frac{1}{n}\right)^n$. Show that the sequence $\{e_n\}_{n=1}^{\infty}$ is increasing.
Hint: Use the Binomial Theorem $(1+a)^n = \sum_{k=0}^{n} \binom{n}{k} a^k$.

**5.** Determine whether the series is convergent or divergent. If it is convergent, find its sum.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n}$

(ii) $\displaystyle\sum_{n=1}^{\infty} \frac{6 \cdot 2^{2n-1}}{3^n}$

(iii) $\displaystyle\sum_{n=1}^{\infty} \frac{e^n}{n^2}$

(iv) $\displaystyle\sum_{n=2}^{\infty} \left( \frac{1}{e^n} + \frac{2}{n^2 - 1} \right)$

(v) $\displaystyle\sum_{n=1}^{\infty} \frac{n^3 + n^2}{n^3 - 2n + 5}$

(vi) $\displaystyle\sum_{n=1}^{\infty} \left( \frac{3}{5^n} + \frac{2}{n} \right)$

**Answer Keys.** **2.** $\frac{3+\sqrt{5}}{2}$ **3.** $\frac{3-\sqrt{5}}{2}$ **5.** (i) $\frac{1}{7}$ (ii) divergent (iii) divergent (iv) $\frac{1}{e(e-1)} + \frac{3}{2}$ (v) divergent (vi) divergent